ICRA 2027 Workshop and Tutorial Proposal

Proposal Title:
Embodied Dexterity for Multimodal Deformable Object Manipulation

Type: ☐ Workshop ☐ Tutorial
Duration: ☐ Full Day ☐ Half Day
Primary Contact: Alessio Caporali, Università di Bologna, alessio.caporali@unibo.it

Keywords:
deformable object manipulation, soft object manipulation, embodied dexterity, contact-rich manipulation, proprioceptive sensing, tactile sensing, compliant and dexterous robotics, multimodal perception, robot learning, vision-language-action models, sim-to-real transfer;

Meeting open points:

- Ask each invited speaker for a tentative title or high-level topic to help refine the workshop scope and ensure a coherent program.
- Add a 6th speaker about foundation models (possibly related to deformables)?
- Define the schedule structure
- WS website? Should be live before Nov 2? + Social
- Proposal plan: final draft 26/11 -> review for all -> submission
- WS award and sponsor

Todo

- Check TC for approval and sponsor (travel award/reimbursement for student best poster), check companies from last year (e.g. prensilia,) [Alessio], Robotextile GmbH [Roman]
- Website [Kevin]
- 6 Speakers, Edward Johns [Roman]
- Fix Schedule

Marcer -> contact Yunzhu Li (on wait!)

0. Response to Pre-proposal Feedback (1 pages max)

Importance and relevance of the proposed topics with respect to the conference scope:

[+] We find a genuine and well-argued case for timeliness in bringing sensing, control, and learning together around deformable-object manipulation.

Thank you for recognizing the workshop's relevance and timeliness.

[-] Spanning hardware, perception, simulation, planning, control, and learning in a single half-day is ambitious. We would encourage a more targeted subset of this scope to ensure deeper engagement in the time available.

We agree that a more focused scope would enable deeper discussion within the half-day format. We will refine the program around a targeted subset of closely connected themes, with other topics providing supporting context. This will allow more time for discussion and meaningful interaction among speakers and participants.

[+/-] Many current foundation models already integrate several of these pieces by design. To sharpen the discussion, we would encourage including voices from that community — whether industry or academic — in the room to provide a productive counterpoint.

We will explicitly welcome foundation-model contributions and seek an additional invited speaker from this community or industry.

Soundness of the plans to engage and foster active participation of the attendees throughout the event:

[+/-] The proposal states the format is designed around active participation rather than lectures, but this is difficult to fully evaluate given the limited space of the pre-proposal. We look forward to seeing this substantiated with more concrete mechanics in the full proposal.

Section 4 now specifies dedicated Q&A, interactive posters, audience polls, and an anonymous question channel.

Inclusiveness of the event to people with different backgrounds and expertise:

[+] The organizers' committed outreach to affinity groups is a genuine strength. We would encourage carrying this same intentionality into the speaker slate to fully deliver on this promise.

We are extending our outreach commitment to speaker and reviewer selection, seeking diversity in expertise, career stage, geography, and gender.

Experience of the organizers:

[+] The organizing team spans institutions across Italy, Spain, the USA, Japan, Sweden, and Korea, with substantial named prior ICRA and IROS workshop experience. This breadth is a real strength for drawing a genuinely international audience.

Thank you for recognizing the team's experience and international breadth.

Overall: We encourage this proposal. Our main ask for the full proposal is to narrow the half day's scope to a more targeted subset of the stated topics, and to bring in industry or foundation-model voices to provide counterpoint in the discussion.

Thank you for the encouragement and the opportunity to develop the full proposal. We will refine the half-day scope and seek industry or foundation-model perspectives to enrich the discussion.

1. Motivation, Scope, and Objectives (1/2 pages max)

Manipulating cloth, cables, food, and tissue is essential for robotics in manufacturing, logistics, healthcare, and domestic environments. Their evolving geometry, contact conditions, and material behavior make state partially observable and require continuous adaptation. This workshop will focus on …..

The topic is timely as foundation models and vision-language-action systems broaden robot capabilities, while tactile sensing, proprioception, and compliant actuation provide richer physical feedback. Connecting these advances raises three questions: Which states, including tension, slip, and contact, must be inferred beyond visual geometry? How can physical models and learned representations support adaptation and recovery? How should robustness across objects, tasks, and embodiments be evaluated? These questions directly engage ICRA's sensing, control, and learning communities, with hardware and simulation providing supporting context.

Building on the ICRA 2024 and 2025 deformable-object workshops and CoRL 2025 RINO, this event emphasizes the connection between multimodal state estimation and closed-loop action. Expected outcomes include shared research priorities, recommendations for evaluating robustness and recovery, and opportunities for collaborative experimental resources.

The primary RAS TC alignment is Robotic Hands, Grasping and Manipulation, through sensing and sensor-based control. Complementary links are Robot Learning, for adaptive policies and transferable representations, and Soft Robotics, for compliant embodiment and sensing.

2. Organizers

Alessio Caporali (University of Bologna, Italy); Kevin Galassi (Eurecat, Spain); Holly Dinkel (Samsung Research, USA); Roman Mykhailyshyn (AIST, Japan); Marcel Büsching (KTH, Sweden); Daehyung Park (KAIST, Republic of Korea); Alberta Longhini (Stanford University, USA); Irene Garcia-Camacho (CSIC-UPC, Spain).

The organizers have substantial experience organizing international robotics events, including the CoRL 2025 workshop "Beyond Rigid Worlds: Representing and Interacting with Non-Rigid Objects (RINO)" and the RSS 2026 "Lab-to-Production Workshop: Toward Industrial-Grade Perception and Manipulation." Alberta Longhini additionally co-organized the IROS 2026 "Deformable Objects Manipulation: Research Foundations, Reproducibility and Real-World Challenges", the ICRA 2026 "LeHome Challenge: 1st Simulation-Driven Competition on Deformable Object Manipulation," the ICRA 2025 workshop "Reflections on Representations and Manipulating Deformable Objects," and the ICRA 2024 workshop "Representing and Manipulating Deformable Objects." Holly Dinkel co-organized the ICRA 2026 "Intelligent Space Robotics and Systems" workshop and the CoRL 2026 "Do Robots Need World Models?"

workshop. Roman Mykhailyshyn additionally co-organized the IROS 2026 “Industrial Applications of Robot Learning”. Daehyung Park additionally co-organized the RSS 2023 “Experiment-oriented Locomotion and Manipulation Research” workshop, the Humanoids 2024 “Designing Interactive Humanoids”. Irene Garcia-Camacho was the main organizer of the 1st and 2nd editions of the “Cloth Manipulation and Perception Competition” at IROS 2022 and ICRA 2023, co-organized the ”8th Robotic Grasping and Manipulation Competition” and the IROS 2022 Workshop on “Robotic Manipulation of Deformable Objects: Challenges in perception, planning and control for soft interaction”.

The team combines expertise in deformable-object perception, modeling, planning, control, and learning, with work spanning across several applications including cloth, ropes, cables, knotting, soft materials, and single-arm, bimanual, and multifingered systems. Their publications and workshop experience, together with academic and industrial perspectives and geographic, sectoral, and gender diversity, support a rigorous and inclusive program.

3. Program and Format (1/2 pages max)

The half-day workshop is designed around active participation rather than lectures. It will open with an interdisciplinary introduction, followed by invited talks offering perspectives from hardware, sensing, modeling, control, and learning. Each will include Q&A. Speakers will be selected to reflect diversity in approach, application domain, career stage, geography, gender, and academic or industrial background. An open call will invite short extended abstracts, selected through double-blind peer review based on relevance, technical quality, clarity, and discussion potential. Up to 15 contributions will be accommodated as posters, with a subset selected for spotlight presentations. Participation and inclusiveness mechanisms are detailed in Section 4.

Time	Activity	Speaker/Moderator/Lead
8-30 - 8:40AM (10 min)	Opening Remarks	Organizers
8:40 - 9:40 AM (60 min)	Invited speaker 1,2,3	
9:40 - 10:00 AM (20 min)	Post oral presentation	
10:00 - 10:45 AM (45 min)	Coffee break & posters	
10:45 - 11:25 (40 min)	Invited Speaker 3,4	
11:25 - 11:45 AM (20 min)	Post oral presentation	
11:45 - 12:20 AM (35 min)	poster session 2	
12:20 - 12:30 (10 min)	Closing remarks	Organizers

Name	Email	Affiliation	Topics (tentative)	Attendance
Manuel Catalano	manuel.catalano@iit.it	Italian Institute of Technology	Compliant hands and grasping	Confirmed
Kazuhiro Kosuge	kkosuge@cityu.edu.hk	University of Hong Kong	Textile manipulation	Confirmed
Tapomayukh Bhattacharjee	tapomayukh@cornell.edu	Cornell University	Assistive manipulation of food and cloth	Confirmed
David Navarro-Alarcon	dnavar@polyu.edu.hk	Hong Kong PolyU	Perception and control for deformable-object manipulation	Confirmed
Uksang Yoo	uksang@berkeley.edu	UC Berkeley	Mechanics-informed learning for DOM	Confirmed
Additional speaker	TBD	TBD	Foundation models for deformable-object manipulation	To be confirmed

The half-day workshop is designed around active participation rather than lectures. It will open with an interdisciplinary introduction, followed by invited talks offering perspectives from hardware, sensing, modeling, control, and learning. Each will include Q&A. Speakers will be selected to reflect diversity in approach, application domain, career stage, geography, gender, and academic or industrial background.

Poster session and call for contributions: Contributions will be solicited through an open call for short abstracts, spotlight talks, and posters, with encouragement for students, early-career researchers, underrepresented groups, and work from institutions or regions less frequently represented at robotics conferences. Submissions will undergo double-blind peer review based on relevance, technical quality, clarity, and potential to stimulate discussion. Accepted contributions will be presented through lightning talks followed by an interactive poster session, giving early-career researchers visibility. Prerecorded 3-minute teasers may help keep the program on schedule. Presenters will also be encouraged to address limitations, failure cases, or unresolved questions alongside successful results to present new research opportunities.

Tentative schedule

Possible idea 1: 4 speakers x 20 minutes each, two poster sessions (before coffee break, and before lunch)

Time	Activity	Speaker/Moderator/Lead
8-50 - 9:00AM (10 min)	Opening Remarks	Organizers
9:00 - 9:40 AM (40 min)	Invited speaker 1,2	
9:40 - 10:00 AM (20 min)	Post oral presentation	
10:00 - 10:45 AM (45 min)	Coffe break & poster session	
10:45 - 11:25 (40 min)	Invited Speaker 3,4	Tapomayukh Bhattacharjee
11:25 - 11:45 AM (20 min)	Post oral presentation	
11:45 - 12:20 AM (35 min)	poster session 2	
12:20 - 12:30 (10 min)	Closing remarks	Organizers

Program alternative

Possible idea 2: 4 speakers x 30 minutes each, poster session (before coffee break, 5 min presentation or video playback)

Time	Activity
08:45 - 09:00 (15 min)	Workshop opening
09:00 - 09:30 (30 min)	Speaker 1
09:30 - 10:15 (45 min)	Spotlights Talks
10:15 - 10:45 (30 min)	Coffee break + Poster presentation
10:45 - 12:15 (90 min)	Speaker 2, 3, 4
12:15 - 12:30 (15 min)	Closing remarks + extra Poster time through the lunch break

Invited Speakers

4. Participant Engagement and Inclusiveness

Active participation and interaction. The workshop will combine invited talks, contributed spotlights, an interactive poster session, and a moderated discussion around shared challenges in deformable-object manipulation. Each invited talk will include dedicated Q&A, and speakers and contributors will be encouraged to present a failure case, limitation, or unresolved question alongside their results. This will provide concrete starting points for discussion across sensing, control, and learning perspectives. Brief audience polls and an anonymous digital question channel will allow attendees to shape the discussion, including those less comfortable speaking publicly.

Contributions and inclusive representation. An open call for short extended abstracts will invite research contributions and preliminary results. We will particularly encourage submissions from students, early-career researchers, underrepresented groups, and institutions or regions less frequently represented at major robotics conferences. Submissions will undergo double-blind peer review based on relevance, technical quality, clarity, and potential to stimulate discussion. Accepted contributions will receive visibility through spotlight talks and interactive posters, creating opportunities for direct exchange with invited speakers and organizers. We will apply the same intentionality to selecting invited speakers and reviewers, seeking

complementary research backgrounds, career stages, geographic representation, and gender diversity.

Outreach and accessibility. The workshop will be publicized through robotics-worldwide, LinkedIn, Bluesky, organizer institutions, and relevant academic and industrial networks. Outreach will connect communities in deformable-object manipulation, tactile and proprioceptive sensing, compliant robotics, control, and robot learning. We will also seek dissemination through affinity groups such as Women in Robotics and Black in Robotics. Announcements will include the call for contributions, deadline reminders, invited speakers, and the final program. We intend to seek collaboration with the RAS Technical Committee on Robotic Hands, Grasping and Manipulation, with complementary outreach to the Robot Learning and Soft Robotics TCs, particularly for publicity and community engagement. Where authors and speakers grant permission, abstracts, presentation materials, and recordings will be made publicly available through the workshop website. A post-workshop summary will share the main conclusions and open questions, extending access to researchers unable to attend and supporting continued exchange.

5. Previous Events and Additional Information

This is a new workshop, independent of previous workshop series. It builds on the organizers' experience with related events, bringing together complementary communities around deformable-object manipulation.

Complementing the engagement and dissemination plans described in Section 4, a dedicated website will be maintained for at least five years after the event. It will archive the program, accepted abstracts, presentation and poster materials, and recordings where permission is granted. A post-workshop summary will consolidate the main conclusions, open challenges, and research priorities identified by participants, providing a lasting resource for the community and supporting continued collaboration.

6. Organizer Requests

We anticipate 30–50 in-person attendees and request a room with standard theater-style seating, together with space for an interactive poster session accommodating approximately 10–15 posters. Poster boards or easels positioned near the meeting room would facilitate discussion during the coffee break. We request standard audiovisual equipment, including a projector and screen, HDMI/USB-C connectivity, a presentation clicker, and a lapel or handheld microphone. An additional roaming microphone would facilitate questions and interaction with the audience. Access to power outlets or power strips near the presentation area would also be helpful.

References (optional)

Include references only where necessary.

Possible schedule (4h)

- Speaker + poster session + Speaker
- Speaker panel + poster session + Speaker panel
- Speakers + poster + speaker + poster
- Speaker + panel + Speaker + panel + Selected Poster (3x5min)

10:30 - 11 expected coffe break

Time	Activity	Speaker/Moderator/Lead
8.30-8.40 (10 min)	Opening Remarks	Organizers
8.40-9.40 (60 min)	Invited speaker 1,2,3	
9.40-10.00 (20 min)	panel 1	
10.00-10.20 (20 min)	poster video presentation	
10.20-11.05 (45 min)	Coffee break & posters	
11:05 - 12:05 (60 min)	Invited Speaker 4,5,6	
12:05 - 12.25 (20 min)	Panel 2	
12:25 - 12:30 (5 min)	Closing remarks	Organizers